

Nazim Belabbaci

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Summary

Innovative Research Engineer specializing in IoT and AI for biomedical applications. Proven expertise in developing wearable technologies, conducting clinical research, and applying machine learning to enhance diagnostic and monitoring solutions. Experienced in full-stack hardware-software integration and advanced prototyping, with a growing portfolio of peer-reviewed publications and patent-pending work. Proficient in key programming languages and platforms including Java, C, Python, Embedded C, Android, and Arduino. Recognized for rapid learning, comfortable leading a team as serving as a team member. Demonstrated success in managing clinical data collection, collaborating with interdisciplinary teams. Seeking a postdoctoral role to advance AI-based healthcare systems, wearable sensing, and remote patient diagnostics in real-world settings.

Research & Academic Experience

Graduate Research Assistant, *CUBICS Lab, University of Massachusetts Lowell*, Lowell, MA | 2022–Present

♦ Created a 1D U-Net-based conditional diffusion model to generate synthetic FTIR spectra for cancer detection; boosted accuracy of ML classifiers by up to 4% over the baseline. Manuscript in preparation; code and trained models available at github.com/NazimBL/SpectralDiffusionLab.

♦ PhD dissertation: leading a controlled exercise hydration study. Drafted and revised the IRB protocol, collecting pre-/post-workout wearable and saliva ground-truth data, performing data cleaning.

♦ Developed Transformer models for real-time classification of fatigue, mood, and activity from wearable time-series physiological signals. Published in MobiQuitous 2024, AMIA 2025; achieved 3 to 10% improvement over previous benchmarks.

♦ Designed a microcontroller-driven wearable for tracking human hydration levels. The prototype uses an integrated light sensor to measure hydration on par with industry-standard laboratory spectrophotometers. Invention Disclosure for a Patent is pending, titled "Methods for Body Hydration Estimation Using Wearable Sensors".

PhD Research Intern, *Beth Israel Deaconess Medical Center (BIDMC)*, Boston, MA | 2025–Present

♦ Responsibilities include analyzing RNA sequencing datasets from five infection models to characterize splicing events in the context of viral infection. Performing computational and AI analysis to identify and characterize events associated with different viral infection stages.

Graduate Teaching Assistant, *University of Massachusetts Lowell*, Lowell, MA | 2023–Present

♦ COMP 2030 Assembly Language & COMP 4200 Artificial Intelligence: Led labs and recitations for 100+ students; graded assignments/exams; held office hours and created supplemental materials.

Graduate Research Assistant, *Takaku Lab, University of North Dakota*, Grand Forks, ND | 2022

♦ Designed interpretable ML models to identify breast cancer-linked DNA binding patterns using ATAC-seq data. ♦ Collaborated with bioinformatics specialist and a web developer to create a user-friendly online platform (MotifXplorer) enabling non-technical clinicians to explore model outputs and regulatory site predictions.

Prior Technical Experience (Algeria)

Co-Founder & Embedded Systems Engineer, Deadline Technologies, Bouira, DZ | 2020–2021

- ◇ Designed, built, and sold wireless-controlled embedded boards for academic and industrial use.
- ◇ Managed full product lifecycle: PCB design, firmware, marketing, and direct sales.

Embedded Software Engineer, EURL Microtechnologies Lab, Boumerdès, DZ | 2019–2020

- ◇ Developed software for laboratory-grade electrical measurement tools.
- ◇ Defined and tested core firmware features for national educational institutions.

Education

PhD, Computer Science, University of Massachusetts Lowell, Lowell, MA | 2023–Present

(Thesis: *“From Light to Insight: Spectroscopy-Driven Hydration Assessment with Embedded AI and Synthetic Data Augmentation”*) (GPA 3.98)

Master, Computer Science, University of Massachusetts Lowell, Lowell, MA | 2023

(Thesis: *“A Prototype Smartwatch for Real-Time Monitoring of Hydration Status”*) (GPA 3.98)

Master of Computer Engineering, National Institute of E&E Engineering, Boumerdès, DZ | 2019

(Thesis: *“Low-cost micro-controller-based Phasor Measurement Unit”*) (GPA 3.7)

Bachelor in E&E Engineering, National Institute of E&E Engineering, Boumerdès, DZ | 2017

(Thesis: *“Eco-Friendly Voice-Controlled Smart Home”*) (GPA 3.7)

Awards & Honors

Finalist of the \$200K Challenge, an annual start-up competition hosted by **M2D2** |2024.

LaTorre Family Scholarship Endowment fund award winner, UML endowed |2023.

Finalist of the UND Middleton School of Entrepreneurship Business Plan Competition |2022

◇ Developed and pitched a 30-page AI-based COVID-19 early-detection business plan using wearable sensor data. Conducted market analysis and financial modeling for the proposed venture.

Winner of Leapfrog International Hackathon of Algiers |2018

◇ Led a 48-hour team sprint to develop “eHealth” an IoT medical monitoring prototype and presented it to a panel of industry experts.

Civic Engagement

Chair, Student’s Initiative, AASTE, Pembroke Pines, FL |2022-2024

- ◇ Led a team of 20 to deliver online STEM webinars to 200+ students.

President & Head Project Manager, *Wameedh Scientific Club*, Boumerdès, DZ |2016-2019

- ◇ Secured funding and ran hands-on workshops and online classes for 100+ student.